

DALIAN RIQIAN MOTOR: SPECIALIZATION OR DIVERSIFICATION?

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Dalian RiQian Motor Co. Ltd. (RiQian Motor) was a manufacturing company located in Dalian City, Liaoning Province, Northeast China. Established in 2003, RiQian Motor was committed to the maintenance, parts manufacturing, and whole machine matching of electric wheel motors for mining machinery. After investing more than ten years of development in this field, RiQian Motor became the leading manufacturing enterprise in this subdivision in China.

Over time, RiQian Motor's high quality products and service won the trust and favour of construction machinery manufacturers. Experienced in everything from electric wheel maintenance to imitation to independent research and development (R&D), RiQian Motor became an expert in the field of electric wheel motors for mining machinery. In 2012, the mining machinery manufacturing industry turned its focus to the development of electric vehicles. RiQian Motor tried to diversify in the direction of this development and entered the field of new energy motors, successfully supplying motors for Dongfeng Motor Corporation Ltd. (Dongfeng)'s new energy mining trucks in 2016. However, as of 2018, RiQian Motor's new energy motor business was still in a loss-making state and needed continuous subsidies from the traditional electric wheel motor business.

In the face of RiQian Motor's financial situation, Chairman Pan Xiaohua (Pan) had to decide which direction the company should take. Those within the company comprised two opposing views: some thought that the traditional electric wheel motor market demand was still large and that RiQian Motor should continue to specialize in that market. Others believed that compared with electric wheel motors, the development of new energy motors was more in line with industry trends, and that RiQian Motor should diversify and enter the new energy motor market. What should Pan do?

THE MINING MACHINERY ELECTRIC WHEEL MOTOR MARKET IN CHINA

High Specialization in Product Manufacturing

The mining machinery electric wheel motor was composed of two parts: the wheel side motor and the power generation motor. The vehicle's power, transmission, and braking system was integrated into a set of motors, which were mounted on the edge of the wheel to drive the wheel separately.

Mining machinery electric wheel motors were used in machinery such as mining dump trucks and wide-body minecarts. These vehicles had a large tonnage, low energy consumption, and high technical requirements. They were used to transport coal, sand, and gravel, usually working in conjunction with construction machinery such as excavators, loaders, and belt conveyors to form production lines for loading, transportation, and unloading. Due to the particularity of mining operations, mining machinery electric wheel motors were usually manufactured in complex processes that had highly specialized technical requirements.

Supply Chain

The mining machinery electric wheel motor was in the middle of the entire industrial chain, and its upstream products were common metallurgical semi-finished products such as castings and bearings. The production process of these products was simple, and there were established suppliers in China, so motor manufacturers did not have to worry about the supply. The market for this was determined by the downstream mining machinery industry market.

In the 1980s, the development of Chinese mining construction machinery was mainly based on imported products. At that time, enterprises imported a large amount of foreign equipment that was manufactured using advanced techniques. In 2000, the infrastructure wave in China started, and ongoing investment in infrastructure construction and real estate construction brought great development opportunities to the mining machinery industry. At this stage, the domestic mining machinery enterprises began to learn foreign advanced technology and developed this technology for themselves. The domestic mining machinery wheel motor market began to develop.¹

Opportunity in the Domestic Market

Until the beginning of the twenty-first century, China's mining machinery electric wheel motor market was dominated by foreign enterprises such as Siemens AG and General Electric Company (GE).² Until the Chinese mining enterprises mastered key technologies themselves, the need to import expensive supplies and spare parts resulted in high costs. For example, the electric wheels of mining dump trucks used in China were still mainly imported until 2008.

In 2006, the chief administrative authority in China, the State Council, promulgated and implemented its *Several Opinions of the State Council on Accelerating the Revitalization of the Equipment Manufacturing Industry*.³ After that, the mining machinery electric wheel motor began to officially move toward independent design.

In the early stage of localization, only large-scale state-owned enterprises were able to produce electric wheel motors because few small- and medium-sized enterprises could meet the technical requirements. However, the market share of this segment was small. The large-scale state-owned enterprises were willing to invest only a small amount in it. Consequently, an embarrassing situation was formed in the market: "small- and medium-sized enterprises could not do it, and large-scale enterprises were unwilling to do it." The market gap was obvious.

¹ "Development History of Mining Machinery in China," [in Chinese], Docin, December 10, 2013, <https://www.docin.com/p-738604315.html>.

² Bian Xie Zu, *History of the Motor Industry Development: A Centennial Review and Outlook*, Chinese edition (Beijing: China Machine Press, 2011).

³ The Central People's Government of the People's Republic of China, *Several Opinions of the State Council on Accelerating the Revitalization of the Equipment Manufacturing Industry*, [in Chinese], June 29, 2006, https://www.gov.cn/jrzq/2006-06/29/content_322286.htm.

THE ESTABLISHMENT OF RIQIAN MOTOR

In 1999, Pan acquired Dalian Traction Motor, an enterprise in Dalian, Liaoning Province, China, and entered the motor industry. In 2003, funded by Dalian Traction Motor and Japan Chiba General Industry Co. Ltd., RiQian Motor was officially established.

In the early stages of operation, due to outdated equipment and a single product line, RiQian motor lacked market competitiveness. However, Pan firmly believed that performance was determined by the market. Through visits to clients, Pan found that foreign motor enterprises couldn't offer service on time, due to limitations caused by geographical distance and time zone differences. Pan recruited a group of engineers from other motor factories and set up a professional motor maintenance team to undertake the maintenance business of a variety of imported motors. This became RiQian Motor. Thanks to the team's rich maintenance knowledge and capacity to solve difficult maintenance problems, RiQian Motor soon made a name for itself in the field of motor repair, especially that of electric wheel motors. The client network prototype had emerged since then.

INITIAL EXPLORATION OF SPECIALIZATION

Based on his in-depth understanding of the domestic electric wheel motor market, Pan asserted that the main force to conquer such market segments must be small- and medium-sized enterprises such as RiQian Motor, which needed to quickly enter the market before other manufacturers. He put forward the company's strategic positioning with the motto "Professional, Refined and Featured" and aimed to become a specialized motor manufacturing service provider.

Over time, the maintenance team of RiQian Motor accumulated rich technical experience from its maintenance work. In Pan's view, for equipment manufacturing enterprises, technology was not only the foundation of the company but also the key to long-term development.

In 2006, after comparing RiQian Motor to competitor Tianyuan Motor and Pump Manufacturing Co. Ltd., Pan recruited a large number of technical specialists and set up an R&D department. Through communication with the maintenance team, the R&D department explored imported product design and quickly accumulated technical knowledge. Tentatively, it began to design and produce components for electric wheel motors, replacing damaged parts of imported products during repairs. After repeated tests, the use of self-produced parts did not cause abnormal operation. RiQian Motor's endeavour won the approval of clients. In 2008, RiQian Motor was distinguished as one of China's Top 100 Regional Economic Innovation Units.⁴

MARKET ORIENTATION: DEVELOPMENT OF THE ELECTRIC WHEEL MOTOR

Pan found that the technological development of RiQian Motor needed to be analyzed from the perspective of the market. He continued to visit new clients, while also maintaining communication with old clients. Through these avenues, he determined that there were opportunities in the electric wheel motor market.

Imported electric wheel motors had monopolized the domestic market for a long time due to their technological advantage. In order to ensure the performance and reliability of the motor, construction machinery clients had to pay for these high-priced imported products. This showed a strong demand for cost-effective domestic motors.

⁴ "RiQian's Qualifications and Honors," [in Chinese], Dalian RiQian Motor Co. Ltd., accessed October 06, 2023, <https://www.dlrq.com/col.jsp?id=114>.

National policy also offered good opportunities for RiQian. In 2010, the State Council issued its *Decision of the State Council on Accelerating the Cultivation and Development of Strategic Emerging Industries*,⁵ which proposed a focus on cultivating and developing high-end equipment manufacturing industries, strengthening introduction, digestion, absorption, and re-innovation, so that China's overall innovation ability and industrial development levels reached the world's advanced level. This policy matched RiQian Motor's strategic positioning well.

Another advantage that Pan identified was that due to the small market share and high requirements for technical specialization, there was no strong domestic competitor in China.

Pan firmly believed that the existing technology monopoly of imported suppliers would not be difficult to crack. RiQian Motor gradually turned its focus to the development of the whole electric wheel motor. In 2010, RiQian Motor organized its R&D department to survey and map GE's electric wheel motor, and to communicate with domestic engineers who understood this product. The R&D team analyzed the parts, design ideas, and assembly methods one by one and absorbed technical knowledge to reverse engineer the assembly of the whole machine from their knowledge of the role of each component part.

RiQian motor also fostered school-enterprise co-operation, working with Dalian University of Technology to develop motor performance calculation software and with Shenyang University of Technology to break through the electromagnetic difficulties of electric wheel motor design. In order to realize the climbing requirements of the minecart, RiQian Motor continuously adjusted the motor torque until its performance was successful. It accomplished this by repeatedly calculating the heat generation, material insulation, and other necessary elements.

Using this method, RiQian Motor successfully absorbed advanced techniques from foreign brand products such as GE, Komatsu Ltd., Caterpillar Inc., Terex Corporation, and BelAZ.⁶ RiQian Motor achieved high product performance, and subsequently became the choice of domestic construction machinery clients for cost-effective and timely service. In 2011, the generator and motor series products produced by RiQian Motor were selected for the *Catalogue of Famous, Excellent, and New Mechanical and Electrical Products* issued by the China Machinery Industry Association.⁷

IN-DEPTH DEVELOPMENT OF SPECIALIZATION

Independent Research and the Development of Ancillary Motors

In 2012, due to a decline in the competitiveness of imitation products, the growth rate of RiQian Motor marketing fell from 28 per cent to -8 per cent (see Exhibit 1). Pan knew that the company's strategy of copying existing technology had to change. His first step was to develop independent ancillary motors.

RiQian Motor undertook the development of an ancillary 120-tonne electric wheel motor for Xuzhou Construction Machinery Group Co. Ltd. (XCMG). This project differed from its previous efforts. Rather than working on an imitation of imported products to achieve local production, RiQian Motor was now required to meet the customized needs of XCMG through R&D.

In order to meet these needs, RiQian Motor integrated internal and external resources, sparing no effort to communicate with XCMG technicians. The project team determined the required performance and the

⁵ The Central People's Government of the People's Republic of China, *Decision of the State Council on Accelerating the Cultivation and Development of Strategic Emerging Industries*, [in Chinese], October 18, 2010, https://www.gov.cn/zwggk/2010-10/18/content_1724848.htm.

⁶ "RiQian's Qualifications and Honors," [in Chinese].

⁷ "RiQian's Qualifications and Honors," [in Chinese].

structure of the motor from the perspective of the whole machine, adjusted the performance, and shaped the design through the model trial installation. They then used the calculation and simulation method to refine the electromagnetic, structural, and fluid components of the internal core technology, designing the custom motor. After that, the project team selected and combined the parts that could match the performance of the whole machine, generated mechanical drawings, and finally produced the physical motor. The final result manufactured by RiQian Motor passed the tests of performance and reliability. The ancillary 120-tonne electric wheel motor was successfully delivered.

The news that RiQian Motor had successfully supplied high-performance and cost-effective products for XCMG spread widely. More and more construction machinery manufacturers put forward ancillary requirements to RiQian Motor for design and manufacture.

Expanding Into the Market of Ancillary Motors and Exploring the Overseas Market

After the success of the XCGM project, Pan expanded into the market of ancillary motors. From 2014 to 2018, RiQian Motor successively completed projects meeting the motor ancillary needs of enterprises including Shenhua Nortel Victory Energy Co Ltd., Sany Heavy Industry Co. Ltd. (Sany), and Lingong Heavy Machinery Co. Ltd (LGMG). Moreover, RiQian Motor provided motor colour customization services based on the needs of clients. Ancillary motors with unique paint such as “XCMG Blue” and “Sany Red” were deeply loved by clients.

Pan’s next step was to explore overseas markets. Affected by the economic downturn, the domestic infrastructure fever had gradually cooled down, and the market growth of the mining machinery industry had slowed as well. By participating in exhibitions in North Korea and Russia, RiQian Motor brought its products to the world stage. During the process of expanding into the international market, RiQian Motor’s previous branding measures played a vital role in bidding for orders. Its dazzling client list, including XCMG, Sany, LGMG, Xiangdian Group Co. Ltd., and Shougang Mining Investment Company Ltd., was like a “golden business card,” becoming the means for RiQian Motor to knock on the door of the international market.⁸

RiQian Motor’s professional approach of “marketing drive plus R&D follow up” succeeded in making the mining machinery electric wheel motor its number one product. Its business scope covered the entire process of electric wheel motor maintenance, parts development, and machine ancillary. In 2016, the growth rate of RiQian Motor’s sales revenue turned from negative to positive, reaching 46 per cent (see Exhibit 1).

NEW ENERGY MOTORS AND THE EXPLORATION OF A DIVERSIFICATION STRATEGY

Since 2012, the State Council had successively issued policies to support and standardize the development of the industrial motor industry, emphasizing that motors should develop in the direction of high efficiency and energy saving.⁹ Due to the influence of low-carbon policies, green energy conservation had increasingly become the focus of RiQian Motor’s major clients and component manufacturers. However, the technical development of new energy motors was very different from the traditional electric wheel motors and other mining machinery motors produced by RiQian Motor. Moving into this market would bring high trial and error costs to the company.

⁸ “The Global Strategic Deployment of Dalian RiQian Motor,” [in Chinese], Dalian RiQian Motor Co. Ltd., accessed October 06, 2023, <https://www.dlrq.com/col.jsp?id=112>.

⁹ “Notice of the State Council on Issuing the Development Plan for Energy Conservation and New Energy Vehicle Industry (2012-2020),” [in Chinese], *Official Gazette of the State Council*, no. 20 (June 28, 2012), https://www.gov.cn/gongbao/content/2012/content_2182749.htm.

Pan realized that electrification of vehicles would be the future of mining industrial motor development. If RiQian Motor could lay out the field in advance and provide ancillary motors for new energy machinery, it could occupy a favourable position in the wave of green transformation of the mining machinery market. Under the leadership of Pan, RiQian Motor invested in new energy machinery motor development, and successfully provided ancillary motors for Dongfeng's hybrid mining vehicle DF45E.¹⁰ RiQian Motor had also become the first manufacturer of large-tonnage new energy mining truck motors in China, and the relevant motor technology had become an industry leader in China.

However, unlike the development of electric wheel motors, the leading advantage of new energy motors had not been quickly translated into financial figures. As of 2018, RiQian Motor's new energy motor business was still in a state of loss.

THE FUTURE OF RIQIAN MOTOR

In the face of high trial and error costs, the company was divided as to which direction to go in. Some thought that the traditional electric wheel motor market demand was still large, and because RiQian Motor's advantage was obvious, it should focus on specializing in that area to stop losses. Others believed that the development of new energy motors was more in line with industry trends and that RiQian Motor should grasp the advantages it had established and continue with its diversification strategy, moving forward in the development and manufacture of new energy motors.

If the company chose to return to specialization, it could continue to deepen the field of traditional mining mechanical motors. There were many advantages to this option: after years of operation, RiQian Motor had mastered the core technology of mining machinery motors, its products were widely recognized by clients, and the market share was in the leading level of the industry. On the other hand, if RiQian Motor chose the strategy of specialization to avoid the risk of investing in the field of new energy motors, it might lose better profit opportunities. However, it could potentially ensure its own financial security and its development might be more stable.

If the company chose diversification, RiQian Motor could rely on the existing technical advantages and transform the exploration of diversification into the company's main development strategy. At the same time, RiQian Motor could also use the existing channel resources to establish a close co-creation relationship with major clients of mining machinery in the field of new energy, quickly layout the market, and transform early exploration into the main growth point of future development. However, diversification would also weaken the market position of RiQian Motor in the traditional electric wheel motor market. The company might invest a lot of resources into the development of new energy motor, but still find that it struggled to profit.

Faced with the opportunities and challenges, how should Pan respond? What strategy was the best option for RiQian Motor?

¹⁰ "Dongfeng Manufacturing the World's First Largest New Energy Mining Vehicle," [in Chinese], Dalian RiQian Motor Co. Ltd., accessed December 10, 2023, <https://www.sohu.com/a/119264773377294>.

EXHIBIT 1: DALIAN RIQIAN MOTOR MAIN OPERATING DATA

Year	Sales revenue (In ¥10 Thousand)	Rate of increase
1999	270	—
2000	420	55%
2001	496	18%
2002	757	53%
2003	1,524	101%
2004	3,914	157%
2005	6,387	63%
2006	10,015	57%
2007	10,030	1%
2008	13,233	32%
2009	13,523	2%
2010	17,231	27%
2011	22,043	28%
2012	20,172	-8%
2013	17,406	-21%
2014	16,021	-8%
2015	12,650	-21%
2016	18,433	46%
2017	23,568	28%
2018	25,516	8%

Note: ¥ = CNY = Chinese yuan renminbi. US\$1.00 = ¥6.86 as of December 28, 2018.

Source: Based on company documents.